Timer Studio

User Guide · Version 2026.09.08.0001

Multi-timer Windows desktop application -- countdown, stopwatch, date/time targets, calendar import, and customisable alerts.

Requirements to Build

Tool	Where
.NET SDK 6 or later	https://dotnet.microsoft.com/download
Windows 10 or 11 (64-bit)	--
NSIS 3.x *(optional, for installer)*	https://nsis.sourceforge.io/

> The compiled exe requires the **.NET 10 Desktop Runtime** for Windows. Download from https://dotnet.microsoft.com/en-us/download/dotnet/10.0

How to Build

```
cd TimerStudio
build.bat
```

Or directly: `dotnet build -c Release`

Output: `bin\Release\net10.0-windows\TimerStudio.exe`

`build.bat` will also run `makensis installer.nsi` automatically if NSIS is on the PATH, producing `TimerStudio_Setup.exe`.

Features

Timer Modes

Mode	Description
Countdown	Set hours / minutes / seconds; alert fires at zero
Stopwatch	Counts up with centisecond precision; paused state persists across restarts
Date / Time	Count down to a specific date and time; supports early reminders

Alerts

Feature	Details
Alert color	16-color palette per timer (8 pastels + 8 vivid)
Alert title color	12-color palette for the name text inside the alert
Alert style	Lens (small floating popup), Fullscreen flash, or Both
Alert sounds	Beep, Chime, Alarm, Bell, Ping, Silent -- with Preview button

Feature	Details
Alert text size	Adjustable font size (12-60 pt) per timer
Auto-dismiss	Configurable timeout (0 = manual dismiss only)
Auto-reset	Restart countdown automatically after each alert
Dismiss / Escape	Click alert popup or press Escape to dismiss

Early Reminders (Date/Time mode)

Feature	Details
Lead time	Set a reminder up to 72 hours before the target
Repeat interval	Repeat the reminder every N minutes between lead time and target
h/m input	Enter hours and minutes directly, or use the slider
VALARM auto-import	Reminder lead time is auto-populated from ICS VALARM data

Calendar Import

Feature	Details
ICS file	Import events from any calendar app (Outlook, Google, Apple, etc.)
Outlook direct	Read upcoming events directly from Outlook desktop via COM
RRULE expansion	Recurring events (daily/weekly/monthly/yearly) are fully expanded
Location field	Event location shown in picker and auto-imported into Notes
VALARM	Pre-event alarm triggers auto-populate the reminder lead time
Multi-select	Select multiple events at once; each creates its own timer card
All-day events	Imported at 9:00 AM on the event date (adjustable after import)

Per-Timer Notes

Each timer card has a collapsible **Notes** panel for free-text notes or descriptions. Calendar imports auto-populate from the event's DESCRIPTION and LOCATION fields. Notes persist across restarts.

Quick Presets

A **Quick preset** dropdown in the alert settings applies common configurations instantly:

Pomodoro (25 min), Short break (5 min), Long break (15 min), 1 hour, End of day (5:00 PM), Midnight.

Window & UX

Feature	Details
Multiple timers	Run unlimited timers simultaneously

Feature	Details
Drag to reorder	Drag a card's header row to change its position
Duplicate timer	button copies all settings to a new card
Collapsible cards	collapse individual cards to save space
Persistent timers	All timers and settings saved automatically, restored on restart
Export / Import	Back up timers to JSON; import on any machine
Always on top	* Pin keeps the window above all other windows
Hide / Show	Hide collapses all cards; alerts still show a banner
Dark / Light theme	Theme toggles between dark and light mode
System tray	Closing minimises to tray; timers keep running
Tray reminders	Balloon tooltip fires when an early reminder triggers
Taskbar progress	Windows taskbar button shows countdown progress and turns red on alert
Start with Windows	Optional startup entry in Windows login sequence
Window position	Position and size persist across restarts; clamped on monitor disconnect
Ctrl+N	Global shortcut to add a new timer

Accessibility

- Full `AutomationProperties.Name` and `HelpText` on every interactive control
- WCAG 1.4.3 AA contrast met by both dark and light themes (body text 7:1)
- WCAG 2.5.5 touch target minimum 44×44px on all position grid buttons
- Keyboard navigation: Escape dismisses alerts and cancels name edits; Enter commits names
- Windows Reduce Motion respected: pulse animation suppressed when animations are off
- Windows High Contrast: automatically switches to light theme when HC is active

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Project Structure

File	Purpose
`App.xaml / .cs`	WPF entry point; theme switching; single-instance guard; High Contrast detection
`MainWindow.xaml / .cs`	Main window; toolbar; tray; global tick; drag-to-reorder; taskbar progress
`TimerCard.xaml / .cs`	UserControl per timer; all state machine, alert, reminder, preset, notes logic
`TimerModel.cs`	Data model: all fields, state machine, display formatting, progress
`TimerPersistence.cs`	Atomic JSON save/load; self-test; stopwatch restore
`AppPrefs.cs`	Window geometry and theme persistence
`LensPopupBase.cs`	Shared base class for all floating alert/reminder popups

File	Purpose
`LensAlertWindow.cs`	Final "Time's up!" lens popup
`LensReminderWindow.cs`	Early reminder lens popup
`FlashOverlayWindow.cs`	Fullscreen alert overlay
`AlertSoundPlayer.cs`	Windows Beep API tones -- zero external audio dependencies
`CalendarImporter.cs`	ICS parser (RFC 5545, RRULE, VALARM, LOCATION) + Outlook COM reader
`CalendarEventPickerDialog.xaml / .cs`	Calendar event picker with checkbox multi-select
`AboutDialog.xaml / .cs`	Version info, startup toggle, Export/Import
`WhatsNewDialog.xaml / .cs`	What's New shown once per version
`Themes/Dark.xaml`	Dark colour tokens (Catppuccin Mocha-inspired)
`Themes/Light.xaml`	Light colour tokens
`app.manifest`	DPI awareness (PerMonitorV2) + UAC
`installer.nsi`	NSIS installer script (requires NSIS 3.x)

Settings Files

File	Location
Timers	`%AppData%\TimerStudio\timers.json`
Preferences	`%AppData%\TimerStudio\prefs.json`
Version seen	`%AppData%\TimerStudio\seen_version.txt`
Timers backup	`%AppData%\TimerStudio\timers.json.bak`

Version

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